Assignment Beta – Question 3
Knowledge distillation, worked by hand

Syed Ukkashah · 23K-0055 · Notebook: Assignment_Beta_Question_no_3_23K-0055.ipynb

Given

Prompt: "The patient was admitted to the ___". Vocabulary: 1 hospital, 2 clinic, 3 ward, 4 school. The correct word is hospital, so y = [1, 0, 0, 0].

	hospital	clinic	ward	school
Teacher logits z_t	5.0	3.0	2.0	0.0
Student logits z_s	3.0	2.5	1.0	0.5

T = 2, α = 0.5, η = 0.5, natural log throughout.

$$L = \alpha \cdot T^2 \cdot KL(p_T \,\|\, q_T) + (1 - \alpha) \cdot CE(y, q_1)$$

$$\text{softmax}(z, T)_i = e\text{^}(z_i/T) \,/\, \Sigma_j\, e\text{^}(z_j/T)$$

All values are rounded to 4 decimal places for display; the arithmetic was done at full precision.

(a) Probabilities at T = 1

```
Teacher:  z/T     = [5.0000, 3.0000, 2.0000, 0.0000]
          e^(z/T) = [148.4132, 20.0855, 7.3891, 1.0000]
          sum     = 176.8878
          p       = [0.8390, 0.1135, 0.0418, 0.0057]

Student:  z/T     = [3.0000, 2.5000, 1.0000, 0.5000]
          e^(z/T) = [20.0855, 12.1825, 2.7183, 1.6487]
          sum     = 36.6350
          p       = [0.5483, 0.3325, 0.0742, 0.0450]
```

T = 1	hospital	clinic	ward	school
Teacher p_1	0.8390	0.1135	0.0418	0.0057
Student q_1	0.5483	0.3325	0.0742	0.0450

At T = 1 the teacher is very sure of hospital (0.839); the student is much less sure (0.548) and gives clinic a third of its mass.

(b) Softened probabilities at T = 2

```
Teacher:  z/T     = [2.5000, 1.5000, 1.0000, 0.0000]
          e^(z/T) = [12.1825, 4.4817, 2.7183, 1.0000]
          sum     = 20.3825
          p       = [0.5977, 0.2199, 0.1334, 0.0491]

Student:  z/T     = [1.5000, 1.2500, 0.5000, 0.2500]
          e^(z/T) = [4.4817, 3.4903, 1.6487, 1.2840]
          sum     = 10.9048
          p       = [0.4110, 0.3201, 0.1512, 0.1177]
```

T = 2	hospital	clinic	ward	school
Teacher p_T	0.5977	0.2199	0.1334	0.0491
Student q_T	0.4110	0.3201	0.1512	0.1177

Dividing by T = 2 flattens both distributions. The teacher's soft targets now show its "dark knowledge": clinic (0.220) and ward (0.133) are plausible, school (0.049) is not.

(c) KL divergence between the softened distributions

$$\mathrm{KL}(p_T \,\|\, q_T) = \Sigma_i \; p_T_{,i} \cdot \ln(p_T_{,i} \,/\, q_T_{,i})$$

```
hospital : 0.5977 · ln(0.5977 / 0.4110) = +0.2239
clinic   : 0.2199 · ln(0.2199 / 0.3201) = -0.0826
ward     : 0.1334 · ln(0.1334 / 0.1512) = -0.0167
school   : 0.0491 · ln(0.0491 / 0.1177) = -0.0430
KL = 0.0816
```

(d) Hard-label cross-entropy

$$\mathrm{CE}(y, q_1) = -\Sigma_i \; y_i \ln q_{1,i} = -\ln q_{1,}\mathrm{hospital}$$

```
CE = -ln(0.5483) = 0.6010
```

(e) Total distillation loss

```
soft part = α · T² · KL = 0.5 · 4 · 0.0816 = 0.1632
hard part = (1 − α) · CE = 0.5 · 0.6010 = 0.3005
L = 0.1632 + 0.3005 = 0.4637
```

(f) Gradient of the loss with respect to the student's logits

Derivation.

- For softmax followed by cross-entropy, ∂/∂z [−Σ y_i ln softmax(z)$_i$] = softmax(z) − y. So ∂CE/∂z_s = q_1 − y.
- KL(p_T ‖ q_T) = Σ p_T ln p_T − Σ p_T ln q_T. The first sum does not depend on z_s. The second is a cross-entropy with soft target p_T, applied to softmax(z_s/T), so the chain rule adds a factor 1/T: ∂KL/∂z_s = (1/T)(q_T − p_T).
- Combining: the T^2 in front of the KL term cancels one 1/T, which is why distillation uses T^2 (it keeps the soft-target gradients on the same scale as the hard ones; Hinton et al., 2015).

$$\partial L/\partial z_s = \alpha \cdot T \cdot (q_T - p_T) + (1 - \alpha) \cdot (q_1 - y)$$

```
q_T − p_T = [-0.1867, 0.1002, 0.0178, 0.0687]
q₁ − y    = [-0.4517, 0.3325, 0.0742, 0.0450]
soft part = 0.5 · 2 · (q_T − p_T) = [-0.1867, 0.1002, 0.0178, 0.0687]
hard part = 0.5 · (q₁ − y)        = [-0.2259, 0.1663, 0.0371, 0.0225]
∂L/∂z_s   = [-0.4126, 0.2665, 0.0549, 0.0912]
```

Checks: the components sum to 0.0000, as they must for any softmax gradient (adding a constant to all logits changes nothing). A central finite-difference gradient (ε = 10^{-6}) agrees with the formula to 2e-10.

Reading the gradient: it is negative only for hospital, so gradient descent raises that logit and lowers the other three. The soft part pushes school down about four times harder than ward (0.069 vs 0.018), because the student over-rates school relative to the teacher (0.118 vs 0.049) while its ward estimate is already close (0.151 vs 0.133). The one-hot label cannot express this.

(g) One gradient-descent step (η = 0.5) and the new loss

$$z_s' = z_s - \eta \cdot \partial L/\partial z_s$$

```
z_s' = [3.0000, 2.5000, 1.0000, 0.5000] − 0.5 · [-0.4126, 0.2665, 0.0549, 0.0912]
     = [3.2063, 2.3668, 0.9725, 0.4544]
```

```
Student T=1:  z/T     = [3.2063, 2.3668, 0.9725, 0.4544]
              e^(z/T) = [24.6873, 10.6629, 2.6446, 1.5752]
              sum     = 39.5701
              p       = [0.6239, 0.2695, 0.0668, 0.0398]

Student T=2:  z/T     = [1.6031, 1.1834, 0.4863, 0.2272]
              e^(z/T) = [4.9686, 3.2654, 1.6262, 1.2551]
              sum     = 11.1154
              p       = [0.4470, 0.2938, 0.1463, 0.1129]
```

$KL(p_T \parallel q_{T'}) = 0.0567$ $\quad CE = -\ln(0.6239) = 0.4718$

$L' = 0.5 \cdot 4 \cdot 0.0567 + 0.5 \cdot 0.4718 = 0.1134 + 0.2359 = 0.3493$

	Logits	q_1 (hospital)	KL	CE	Loss L
Before	[3.0000, 2.5000, 1.0000, 0.5000]	0.5483	0.0816	0.6010	0.4637
After one step	[3.2063, 2.3668, 0.9725, 0.4544]	0.6239	0.0567	0.4718	0.3493

The loss falls from 0.4637 to 0.3493 (−0.1145, 24.7%). Both parts drop: the student's soft distribution moves closer to the teacher's (KL 0.0816 → 0.0567) and its probability for the correct word rises (0.5483 → 0.6239). This confirms the gradient direction is correct.